

Lab 4: A Century and a Half of Campaigns
84-355 Democracy’s Data | Week 4

Your name here
August 11, 2026

Contents

Before we start 1

Part 1: 160 years in one table 2

Part 2: Why we will work in two-party shares 3

 When a missing value is a historical fact 3

Part 3: The longest-running fact in American politics 4

 A dead man wins two states 6

Part 4: How different are the states from one another? 6

Part 5: The vanishing battleground 8

Part 6: What this data cannot tell you 10

Part 7: On your own 10

 Where the data came from 14

Before we start

Chapter 3 is about how American campaigns changed: from party-run operations to candidate-run ones, from conventions to primaries, from newspapers to television to whatever we are in now.

This lab looks at the other half of that story. Campaigns did not transform in a vacuum — they transformed on a **map that was itself transforming**. For most of the period we are about to look at, a third of the country was not contested at all. No amount of television changes a campaign’s behaviour as much as a state becoming winnable.

We have 41 presidential elections, 1864 to 2024, state by state. Everything is base R. Work in order — later chunks use objects made earlier.

Part 1: 160 years in one table

```
pres <- read.csv("data/pres_states_1864_2024.csv", stringsAsFactors = FALSE)
dim(pres)
```

```
#> [1] 1913    7
```

```
head(pres)
```

```
#>   year state_abbrev democrat republican other      winner party_win
#> 1 1864          CA    41.40     58.60   NA Abraham Lincoln republican
#> 2 1864          CT    48.62     51.38   NA Abraham Lincoln republican
#> 3 1864          DE    51.80     48.20   NA George B. McClellan  democrat
#> 4 1864          IA    36.92     63.08   NA Abraham Lincoln republican
#> 5 1864          IL    45.60     54.40   NA Abraham Lincoln republican
#> 6 1864          IN    46.50     53.50   NA Abraham Lincoln republican
```

Nearly two thousand rows: one per state per election. `democrat` and `republican` are percentages of the total vote in that state.

How many elections, and how many states in each?

```
length(unique(pres$year))
```

```
#> [1] 41
```

```
range(pres$year)
```

```
#> [1] 1864 2024
```

```
head(table(pres$year), 6)
```

```
#>
#> 1864 1868 1872 1876 1880 1884
#>  25  33  37  37  38  38
```

```
tail(table(pres$year), 3)
```

```
#>
#> 2016 2020 2024
#>  51  51  51
```

Twenty-five states in 1864, fifty-one now. That is not a data problem — it is the country growing. But it means **every comparison across time is a comparison between different sets of states**, and we will have to keep saying so.

Look again at 1864: twenty-five states. There were thirty-six in the Union that year.

Eleven states are missing from 1864 because they were in armed rebellion. The file does not say so. It just has fewer rows. This is the first thing to understand about historical data: absence is never annotated.

Part 2: Why we will work in two-party shares

Last week's lab found that third-party votes go missing from election files. This file has the same problem, and is honest enough that we can measure it:

```
sum(is.na(pres$other))
```

```
#> [1] 1365
```

```
nrow(pres)
```

```
#> [1] 1913
```

Most rows have no third-party number at all. So `democrat + republican` will not add to 100, and the gap is not meaningful:

```
p24 <- pres[pres$year == 2024, ]  
summary(p24$democrat + p24$republican)
```

```
#>      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.  
#>  95.95  97.37   98.06   97.89   98.47   99.26
```

The fix is the standard one for historical comparison: ignore everyone else and ask what share of the **two-party** vote each side won.

```
pres$d2 <- 100 * pres$democrat / (pres$democrat + pres$republican)  
round(head(pres$d2), 1)
```

```
#> [1] 41.4 48.6 51.8 36.9 45.6 46.5
```

This is a real choice with a real cost. In 1912, 1968 and 1992 — Roosevelt, Wallace, Perot — a two-party share hides most of what happened. Keep that in mind when those years look strange later.

When a missing value is a historical fact

Nineteen rows have no Democratic or no Republican share at all:

```
absent <- pres[is.na(pres$d2), c("year", "state_abbrev", "democrat",  
                                "republican", "winner")]  
absent
```

```
#>      year state_abbrev democrat republican      winner  
#> 250  1892          CO         NA      41.13 James B. Weaver  
#> 253  1892          FL      85.01         NA Grover Cleveland  
#> 256  1892          ID         NA      44.31 James B. Weaver  
#> 259  1892          KS         NA      48.40 James B. Weaver  
#> 272  1892          NE         NA      43.56 Benjamin Harrison  
#> 290  1892          WY         NA      50.52 Benjamin Harrison  
#> 457  1908          OK      47.99         NA William Jennings Bryan
```

```
#> 474 1912      AZ  43.60      NA      Woodrow Wilson
#> 475 1912      CA  41.81      NA      Theodore Roosevelt
#> 479 1912      FL  69.52      NA      Woodrow Wilson
#> 498 1912      NE  44.77      NA      Woodrow Wilson
#> 505 1912      OK  46.95      NA      Woodrow Wilson
#> 510 1912      SD  42.07      NA      Theodore Roosevelt
#> 516 1912      WA  26.90      NA      Theodore Roosevelt
#> 619 1924      CA      NA     57.20     Calvin Coolidge
#> 660 1924      WA      NA     52.20     Calvin Coolidge
#> 661 1924      WI      NA     37.06 Robert M. La Follette
#> 904 1948      AL      NA     19.04      Strom Thurmond
#> 1099 1964     AL      NA     69.45      Barry Goldwater
```

These are not gaps. Each one records a state where a major party **had no line on the ballot**. In 1892 the Populists fused with the Democrats across the mountain west, so there was no separate Democratic ticket. In 1912 the Progressives kept Taft off California's ballot entirely.

And then:

```
absent[absent$state_abbrev == "AL", ]
```

```
#>      year state_abbrev democrat republican      winner
#> 904  1948      AL      NA      19.04  Strom Thurmond
#> 1099 1964      AL      NA      69.45 Barry Goldwater
```

In 1948 and again in 1964, there was no Democratic slate on the Alabama ballot.

In 1948 the Dixiecrats replaced it; in 1964 Alabama Democrats refused to pledge to Johnson. A voter who wanted to vote for Harry Truman, or for Lyndon Johnson, could not do it.

The ballot is not a neutral instrument. Hold on to this — it is the whole subject of Part 3.

Part 3: The longest-running fact in American politics

Define the eleven states of the Confederacy. This is a choice, and you should see it being made:

```
south <- c("AL", "AR", "FL", "GA", "LA", "MS", "NC", "SC", "TN", "TX", "VA")
pres$region <- ifelse(pres$state_abbrev %in% south, "South", "Rest of country")
```

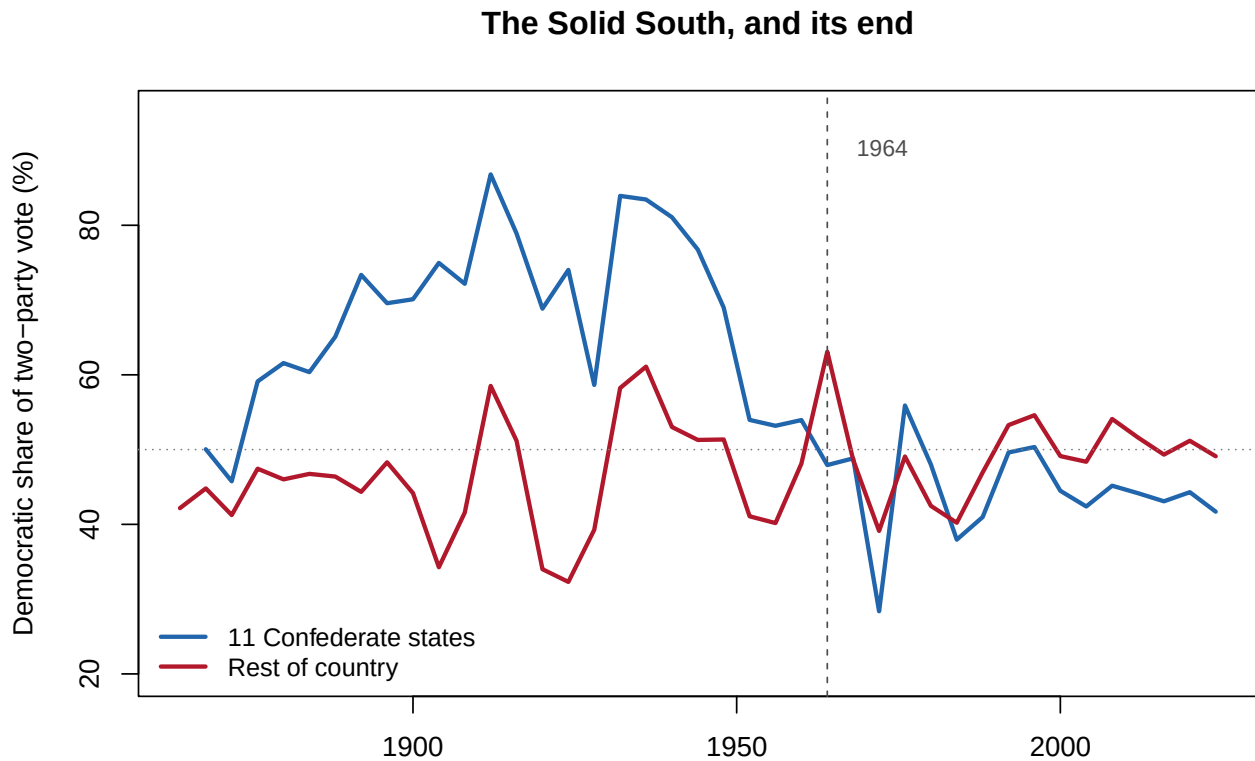
Now average the Democratic two-party share in each region, election by election:

```
reg <- aggregate(d2 ~ year + region, data = pres, FUN = mean, na.rm = TRUE)
wide <- reshape(reg, idvar = "year", timevar = "region", direction = "wide")
names(wide) <- c("year", "rest", "south")
wide$gap <- round(wide$south - wide$rest, 1)
head(wide, 4)
```

```
#>   year    rest    south  gap
#> 1 1864 42.17000      NA   NA
#> 2 1868 44.80962 50.04571  5.2
#> 3 1872 41.24521 45.76188  4.5
#> 4 1876 47.44317 59.11692 11.7
```

Plot both lines:

```
plot(wide$year, wide$south, type = "l", lwd = 2.5, col = "#2166AC",
     ylim = c(20, 95), xlab = "", ylab = "Democratic share of two-party vote (%)",
     main = "The Solid South, and its end")
lines(wide$year, wide$rest, lwd = 2.5, col = "#B2182B")
abline(h = 50, lty = 3, col = "grey50")
abline(v = 1964, lty = 2, col = "grey30")
text(1966, 90, "1964", pos = 4, cex = 0.85, col = "grey30")
legend("bottomleft", legend = c("11 Confederate states", "Rest of country"),
     col = c("#2166AC", "#B2182B"), lwd = 2.5, bty = "n", cex = 0.9)
```



Pause and read that. For eighty years the blue line sits far above the red one — the South voting 70, 80, sometimes 85 percent Democratic while the rest of the country split near even. Then the lines cross, and stay crossed.

When exactly?

```
wide[wide$year >= 1948 & wide$year <= 1976, c("year", "south", "rest", "gap")]
```

```
#>   year  south  rest  gap
#> 22 1948 68.97906 51.35009 17.6
#> 23 1952 53.97186 41.08139 12.9
#> 24 1956 53.18194 40.17258 13.0
#> 25 1960 53.95030 48.12928  5.8
#> 26 1964 47.93240 63.08744 -15.2
#> 27 1968 48.82639 48.61236  0.2
#> 28 1972 28.37745 39.12451 -10.7
#> 29 1976 55.89519 49.07220  6.8
```

The gap peaks around **+42 points in 1924**. It is still nearly +18 in 1948. It goes negative for the first time in **1964** — the year the Civil Rights Act passed, when Johnson won a national landslide and Goldwater carried the Deep South.

It does not vanish overnight. Carter, a Georgian, pulls it back positive in 1976 (+6.8) and it holds in 1980 (+5.5). **From 1984 onward it is negative in every single election**, and by 2024 it is -7.4 .

By 2024 the South is about 7 points *more* Republican than the rest of the country. The sign has flipped and stayed flipped.

A dead man wins two states

While we are here, look at who the file says won in 1960:

```
table(pres$winner[pres$year == 1960])
```

```
#>
#> Harry S. Truman John F. Kennedy  Richard Nixon
#>                2              22             26
```

```
pres[pres$year == 1960 & pres$winner == "Harry S. Truman",
      c("year", "state_abbrev", "democrat", "republican", "other", "winner")]
```

```
#>      year state_abbrev democrat republican other      winner
#> 1049 1960          AL    56.41      42.18 57.43 Harry S. Truman
#> 1072 1960          MS    36.34      24.67 38.99 Harry S. Truman
```

Harry Truman was not a candidate in 1960. He had left office eight years earlier.

What actually happened is more interesting than a typo. Alabama and Mississippi ran slates of **unpledged electors** — Democrats who appeared on the ballot promising *not* to support the national ticket, hoping a close election would let the South bargain over civil rights. Mississippi's unpledged slate won outright. Whoever compiled this file needed a name for that slate and reached for a Democrat, and Truman is what came out.

This is a Week 2 problem wearing 1960 clothes. Somebody made a judgement call while assembling a file, wrote nothing down, and the judgement is now sitting in your data looking like a fact. Notice that it did **not** corrupt the numbers we have been using — the vote shares are fine. Only the label is wrong. Bad data is rarely uniformly bad, which is exactly what makes it dangerous.

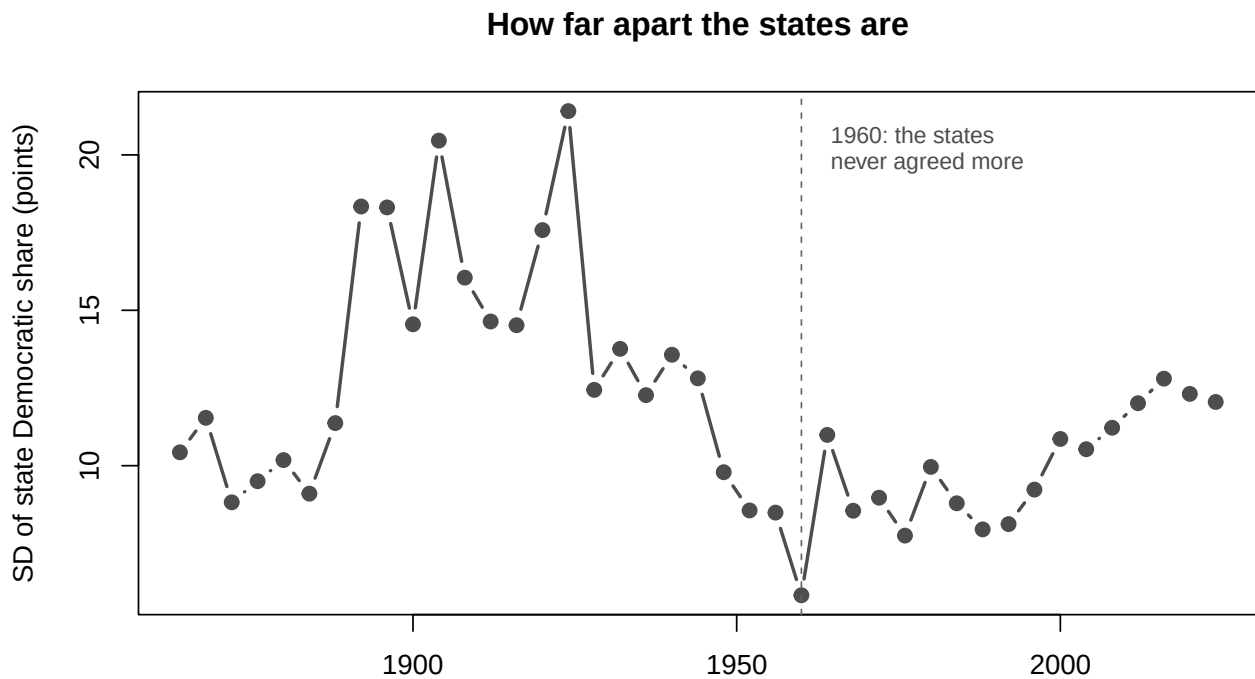
Part 4: How different are the states from one another?

If every state voted alike, the spread between them would be zero. If the country were made of separate political worlds, the spread would be large. Standard deviation across states, election by election:

```
spread <- aggregate(d2 ~ year, data = pres, FUN = sd, na.rm = TRUE)
names(spread)[2] <- "sd"
spread$sd <- round(spread$sd, 2)
tail(spread, 5)
```

```
#>   year   sd
#> 37 2008 11.22
#> 38 2012 12.01
#> 39 2016 12.80
#> 40 2020 12.31
#> 41 2024 12.05
```

```
plot(spread$year, spread$sd, type = "b", pch = 19, lwd = 2, col = "#4D4D4D",
     xlab = "", ylab = "SD of state Democratic share (points)",
     main = "How far apart the states are")
abline(v = 1960, lty = 2, col = "grey40")
text(1962, 20, "1960: the states\nnever agreed more", pos = 4, cex = 0.8,
     col = "grey30")
```



The most-alike year and the least-alike years:

```
head(spread[order(spread$sd), ], 3)
```

```
#>   year   sd
#> 25 1960 5.83
#> 29 1976 7.75
#> 32 1988 7.95
```

```
head(spread[order(-spread$sd), ], 3)
```

```
#>   year   sd
#> 16 1924 21.41
#> 11 1904 20.46
#> 8  1892 18.34
```

The spread was enormous in the era of the Solid South — the South was voting differently from everyone else, and that shows up as huge variance. It collapses to a minimum of **5.8 in 1960**. And then it climbs again, roughly doubling by 2024.

Careful here. That one number is made of parts. Some of the distance between states is a whole region differing from the rest of the country; the rest is states differing from their own neighbours. The next chunk splits the spread into those two pieces.

Before you run it, write two numbers down. In 1924, with the Solid South intact, **what percentage of all the state-to-state variation was simply the South differing from everyone else? And what is that percentage in 2024?** The standard deviations invite you to think the country has gone half the way back, in which case the second number ought to be a solid fraction of the first. Commit to two percentages — not “high” and “lower”.

```
between <- function(y) {
  s <- pres[pres$year == y & !is.na(pres$d2), ]
  g <- s$region == "South"
  gm <- tapply(s$d2, g, mean); nn <- table(g); tot <- mean(s$d2)
  100 * sum(nn * (gm - tot)^2) / sum((s$d2 - tot)^2) # between-region share
}

pb <- sapply(c(1924, 2024), between)
names(pb) <- c("1924", "2024")
round(pb, 1)
```

```
#> 1924 2024
#> 71.8 6.5
```

71.8% in 1924, and 6.5% in 2024. The first number is usually written down too low and the second far too high — most people put 2024 in the thirties or forties, on the reasonable-sounding logic that a spread half the way back ought to be half a regional story. It is not. The states are further apart today than in 1960, but *not* for the 1920s reason. Then, one region was voting unlike the rest. Now 93.5% of the spread is states differing from their own neighbours, everywhere, along lines no regional label captures — this is the sorting you have heard called polarization. The same statistic, two completely different worlds.

A number going up does not tell you why.

Part 5: The vanishing battleground

Here is the fact that matters most for how campaigns behave. Count the states within five points of an even split:

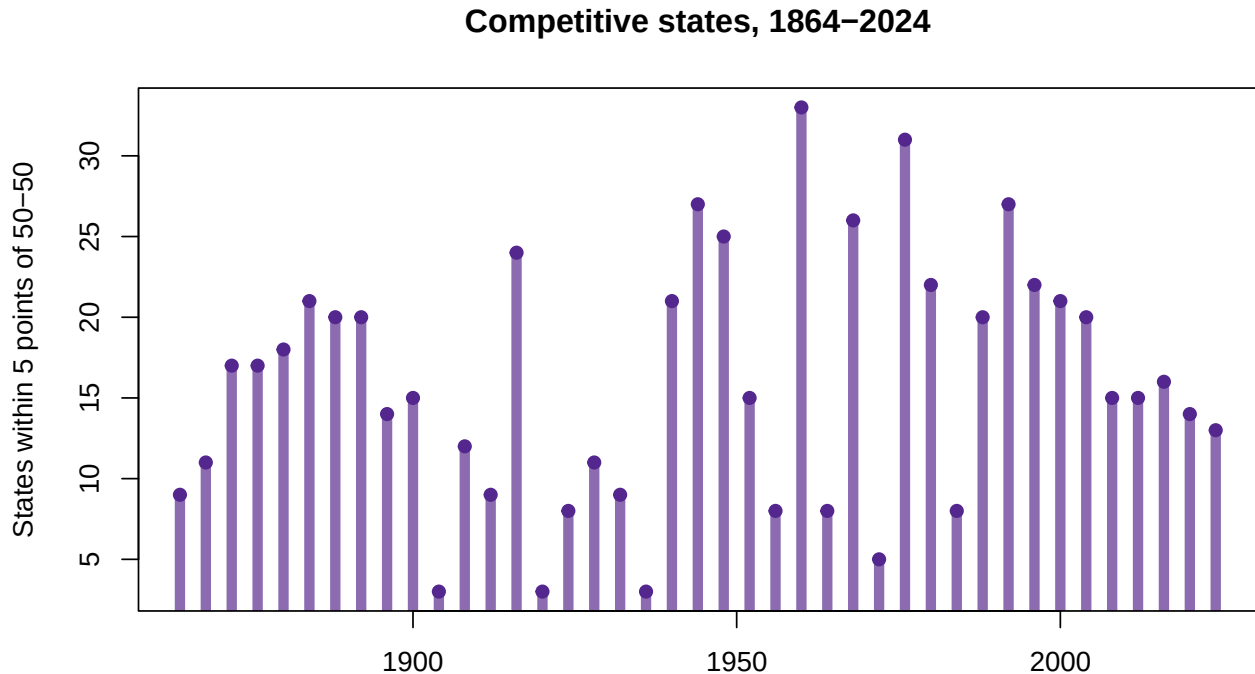
```
pres$close <- abs(pres$d2 - 50) < 5
nclose <- aggregate(close ~ year, data = pres, FUN = sum, na.rm = TRUE)
tail(nclose, 6)
```

```
#>   year close
#> 36 2004    20
```



```
#> 37 2008    15
#> 38 2012    15
#> 39 2016    16
#> 40 2020    14
#> 41 2024    13
```

```
plot(nclose$year, nclose$close, type = "h", lwd = 6, col = "#8C6BB1",
     xlab = "", ylab = "States within 5 points of 50-50",
     main = "Competitive states, 1864-2024")
points(nclose$year, nclose$close, pch = 19, col = "#54278F")
```



```
nclose[nclose$year %in% c(1960, 1976, 2000, 2012, 2020, 2024), ]
```

```
#>   year close
#> 25 1960    33
#> 29 1976    31
#> 35 2000    21
#> 38 2012    15
#> 40 2020    14
#> 41 2024    13
```

Thirty-three competitive states in 1960. Thirteen in 2024 — out of fifty-one, and with far more states in the union than in 1960.

This is the sentence to take away from the lab. A campaign in 1960 had to think about most of the country. A campaign in 2024 has to think about a dozen states, and can safely ignore the rest. Nothing in Chapter 3 about consultants or television explains as much about modern campaign behaviour as this does.

We will see the consequence directly in Week 6, when we look at where candidates actually go.

Part 6: What this data cannot tell you

Three limits, stated plainly, because you should be able to state them yourself by now.

The set of states changes. Comparing the spread in 1864 (25 states) to 2024 (51) is comparing different objects. The trend is real but the early numbers are noisier than they look.

```
range(table(pres$year))
```

```
#> [1] 25 51
```

Two-party share hides third parties. 1912, 1968 and 1992 all look tamer here than they were.

```
pres[pres$year == 1912 & pres$state_abbrev %in% c("CA", "SD", "WA"),  
     c("year", "state_abbrev", "democrat", "republican", "other", "winner")]
```

```
#>      year state_abbrev democrat republican other      winner  
#> 475 1912          CA    41.81           NA 41.83 Theodore Roosevelt  
#> 510 1912          SD    42.07           NA 50.56 Theodore Roosevelt  
#> 516 1912          WA    26.90           NA 35.20 Theodore Roosevelt
```

Theodore Roosevelt won California, South Dakota and Washington in 1912. He does not appear in our d2 column at all — he is inside **other**, or missing entirely.

Shares are not votes. Every number here is a percentage. A five-point margin in Wyoming and a five-point margin in California are the same number and nothing like the same event.

Part 7: On your own

This is the part you turn in. **You are not being asked to write code.** Each option is a working chunk with one thing to change, or nothing at all. Run it, look at it, and tell me what it means.

Pick **one**.

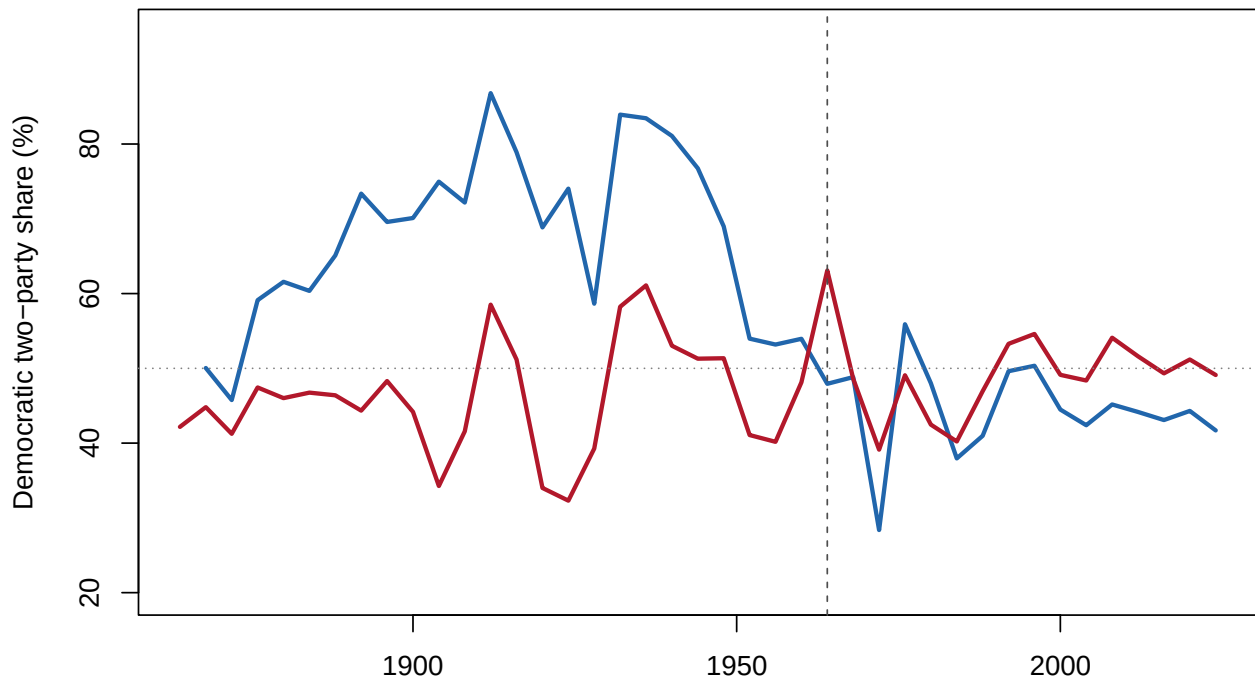
Option A — Move the line

Part 3 used the eleven Confederate states. That was a choice. Here it is again with a different one — the eleven plus the border states that permitted slavery but did not secede.

```
# Change this list and re-run. Try adding "KY", "MO", "MD", "WV", "OK", "DE".  
my_south <- c("AL", "AR", "FL", "GA", "LA", "MS", "NC", "SC", "TN", "TX", "VA")  
  
pres$my_region <- ifelse(pres$state_abbrev %in% my_south, "South", "Rest")  
r <- aggregate(d2 ~ year + my_region, data = pres, FUN = mean, na.rm = TRUE)  
w <- reshape(r, idvar = "year", timevar = "my_region", direction = "wide")  
names(w) <- c("year", "rest", "south")
```

```
plot(w$year, w$south, type = "l", lwd = 2.5, col = "#2166AC", ylim = c(20, 95),
     xlab = "", ylab = "Democratic two-party share (%)",
     main = paste("The South as", length(my_south), "states"))
lines(w$year, w$rest, lwd = 2.5, col = "#B2182B")
abline(h = 50, lty = 3, col = "grey50"); abline(v = 1964, lty = 2, col = "grey30")
```

The South as 11 states



```
w$gap <- round(w$south - w$rest, 1)
w[w$year %in% c(1924, 1948, 1960, 1964, 1976, 2024), ]
```

```
#>   year    rest  south  gap
#> 16 1924 32.30278 74.02989 41.7
#> 22 1948 51.35009 68.97906 17.6
#> 25 1960 48.12928 53.95030  5.8
#> 26 1964 63.08744 47.93240 -15.2
#> 29 1976 49.07220 55.89519  6.8
#> 41 2024 49.10741 41.69793 -7.4
```

Write about: how much does the story depend on which states you call Southern? Does the 1964 crossover move, sharpen, or blur — or hold exactly where it was? Report what you actually find, including “almost nothing changed” if that is the answer.

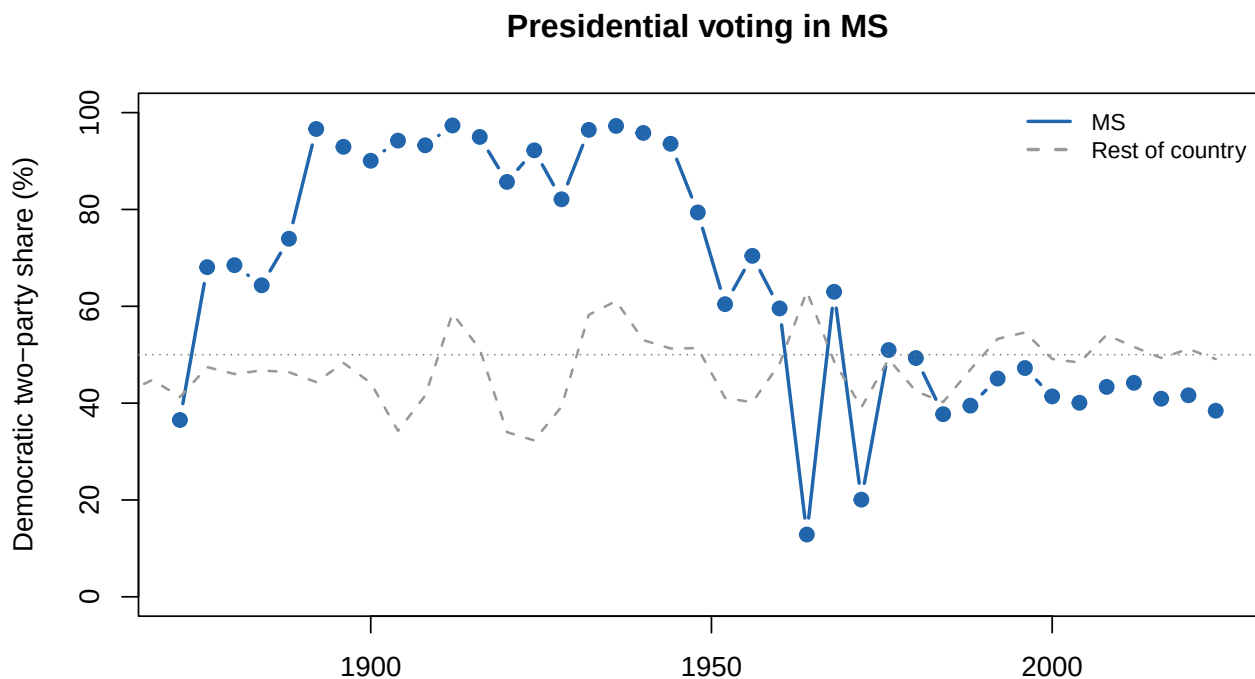
Then the real question, whichever way it went: a finding that survives a change in its own definition is worth more than one that does not. Did this one survive? And what should you now think of anyone — a journalist, a campaign, a professor — who reports “the South” as a single number without telling you which states they meant?

Option B — Pick a state and follow it

```
this_state <- "MS" # <- change to any two-letter abbreviation

s <- pres[pres$state_abbrev == this_state, ]
s <- s[order(s$year), ]

plot(s$year, s$d2, type = "b", pch = 19, lwd = 2, col = "#2166AC",
     ylim = c(0, 100), xlab = "", ylab = "Democratic two-party share (%)",
     main = paste("Presidential voting in", this_state))
abline(h = 50, lty = 3, col = "grey50")
lines(wide$year, wide$rest, lwd = 1.5, col = "grey60", lty = 2)
legend("topright", c(this_state, "Rest of country"),
     col = c("#2166AC", "grey60"), lwd = 2, lty = c(1, 2), bty = "n", cex = 0.85)
```



```
s[s$year %in% c(1948, 1960, 1964, 1972, 2024), c("year", "d2", "winner")]
```

```
#>      year      d2      winner
#> 926  1948 79.38631 Strom Thurmond
#> 1072 1960 59.56401 Harry S. Truman
#> 1123 1964 12.86000 Barry Goldwater
#> 1225 1972 20.06542 Richard Nixon
#> 1888 2024 38.42653 Donald Trump
```

Write about: narrate the line. Where does your state break from the national pattern, and when does it rejoin? Try Mississippi, then try Vermont — which ran the same journey in the opposite direction. Then say what a campaign would have concluded about your state in 1950, and in 2024. Are those the same campaign?

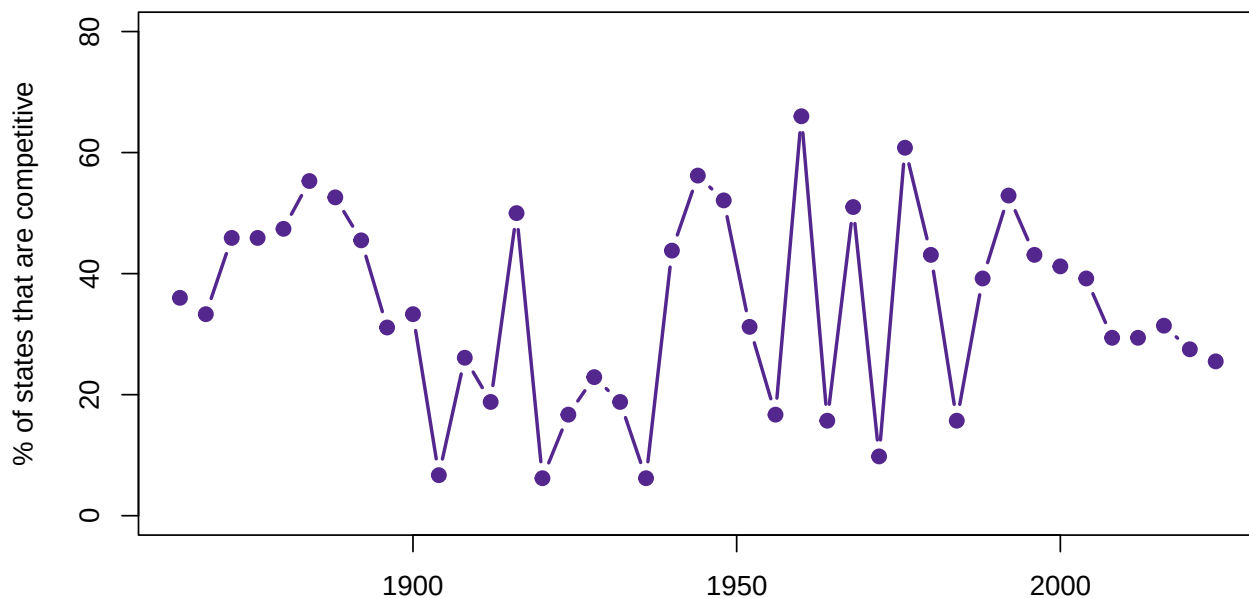
Option C — When did the battleground shrink?

```
margin <- 5 # <- change this threshold and re-run

pres$is_close <- abs(pres$d2 - 50) < margin
nc <- aggregate(is_close ~ year, data = pres, FUN = sum, na.rm = TRUE)
tot <- as.data.frame(table(pres$year)); names(tot) <- c("year", "n")
nc$year <- as.numeric(nc$year); tot$year <- as.numeric(as.character(tot$year))
nc <- merge(nc, tot, by = "year")
nc$pct <- round(100 * nc$is_close / nc$n, 1)

plot(nc$year, nc$pct, type = "b", pch = 19, lwd = 2, col = "#54278F",
      ylim = c(0, 80), xlab = "", ylab = "% of states that are competitive",
      main = paste("States within", margin, "points of even"))
```

States within 5 points of even



```
nc[nc$year %in% c(1960, 1976, 1992, 2008, 2024), ]
```

```
#>   year is_close  n  pct
#> 25 1960      33 50 66.0
#> 29 1976      31 51 60.8
#> 33 1992      27 51 52.9
#> 37 2008      15 51 29.4
#> 41 2024      13 51 25.5
```

Write about: this version uses the *share* of states rather than the count, which is the fairer comparison when the number of states keeps changing. Does the decline still hold? Now change `margin` to 2 and to 10 — does the story survive all three thresholds, or only some? A finding that appears at one cut-off and vanishes at another is worth less, and you should say so if that is what you see.

Option D — Ask your own question

Something in here is bound to look wrong. 1912 and 1968 are strange for real reasons. Vermont and Mississippi swapped places. Follow whatever nags at you. Re-running an existing chunk with a different state or year is plenty.

What to submit. Knit to HTML or PDF and upload to Canvas, with your chosen chunk and **two or three paragraphs** of interpretation.

The writing is the assignment. I am not grading your code — it is my code. I am grading whether you can look at 160 years of election results and say something true about how campaigning changed.

Where the data came from

State returns, 1864–2024 — Jason Timm, *PresElectionResults* (<https://github.com/jaytimm/PresElectionResults>), state series compiled from per-election sources resting ultimately on Dave Leip’s *Atlas of U.S. Presidential Elections* and the official state canvasses.

National popular and electoral vote — the same compilation, in `data/pres_national.csv`. Not used above; it is there for Option D.

Two warnings the source deserves. Nineteenth-century “popular vote” totals rest on state records of wildly varying quality, and in much of the South after 1877 they rest on elections conducted under systematic disenfranchisement — a 90 percent Democratic share in 1900 Mississippi is an accurate count of a vote that most adults in Mississippi were prevented from casting. The number is real. What it measures is not what it appears to measure.

The script that downloaded, checked and wrote these files is `data/build-data.R`. It prints its own sanity checks, including the nineteen rows where a major party was absent from the ballot.